Supplementary Notes for "quantmsdiann: a scalable SDRF-driven DIA-NN workflow for reanalysis of single-cell, spatial, and bulk proteomics datasets"

Contents

Supplementary Note 1. Datasets benchmarked and reanalysed

Table S1 lists every public dataset used in this study. For each dataset, an SDRF was prepared from the original sample metadata and is provided alongside this supplementary deposition. Raw files were downloaded from PRIDE and MassIVE using pridepy [1]. For each cell-line reanalysis cohort, the original published quantification was downloaded from the PRIDE deposition and used as the comparator. The cell-line bulk DIA cohorts were assembled by programmatic search of PRIDE Archive via the PRIDE API [2], filtering for projects annotated as DIA acquisition on human samples with at least one standardised cell line listed in the Cellosaurus reference, downloadable raw files, and SDRF-compatible sample metadata.

Table S1: Datasets benchmarked and reanalysed with `quantmsdiann`. The "Public date" column is the PRIDE/MassIVE public-release year of the deposit, which can predate the associated publication year cited in the main text (e.g. PXD003539 was deposited in 2016 and published as Guo *et al.* 2019; PXD004701 deposited 2016, published as Sun *et al.* 2023).

Accession	**Instrument**	**Type**	**Public date**
ProteoBench benchmarks			
PXD049412	Orbitrap Astral	Single-cell (Module 9)	2024
PXD062685	timsTOF SCP	diaPASEF (Module 5)	2025
PXD070049	ZenoTOF 7600	Mixed-species (Module 10)	2025
Module 7	Orbitrap Astral	Mixed-species (Module 7)	2025
Bulk cell-line panel			
PXD003539	TripleTOF 5600	Cell line (NCI-60)	2016
PXD030304	TripleTOF 6600	Cell line (ProCan)	2022
PXD004701	TripleTOF 5600	Cell line (Sun 2023)	2016
PXD041421	timsTOF (diaPASEF)	Cell line (Wang 2023)	2023
PXD017199	Q Exactive Plus	Cell line (Tognetti 2021)	2021
Single-cell proteomics			
PXD046357	Orbitrap Astral	Single-cell HeLa (Astral)	2024
PXD044991	Orbitrap Astral	Single-cell HeLa (One-Tip)	2024
MSV000093870	Q Exactive	Single-cell oocyte plexDIA (Galatidou 2024)	2024
Phosphoproteomics			
PXD034128	timsTOF Pro	Phospho diaPASEF (Skowronek 2022)	2022
PXD034623	Orbitrap Exploris 480	Phospho directDIA (Galectin-1)	2023
PXD049692	timsTOF HT	Phospho diaPASEF (NK cells)	2024
Spatial proteomics			
PXD064049	timsTOF SCP	Spatial DVP, diaPASEF (MYCN neuroblastoma)	2025
Scalability experiment			
PXD071075	Orbitrap Eclipse	Single-cell	2025

Supplementary Note 2. Building DIA-NN container images

Every quantmsdiann step runs in a versioned container (Docker, Singularity, or Apptainer). Most images are pulled automatically from public registries (BioContainers [3] or ghcr.io/bigbio); DIA-NN is the exception because its license restricts redistribution.

Only DIA-NN 1.8.1 may be redistributed, so it is shipped as the default and pulled from BioContainers (docker.io/biocontainers/diann:v1.8.1_cv1). Releases from 1.9 onward (1.9.2 through 2.5.1, including the 2.5.1-enterprise build) cannot be redistributed and are therefore *not* shipped as images; instead, the companion quantms-containers repository (https://github.com/bigbio/quantms-containers) provides one build recipe per release.

A user with a valid DIA-NN download builds the image locally from the matching recipe, for example:

```
cd diann-2.5.1 && docker build -t ghcr.io/bigbio/diann:2.5.1 .
```

(the equivalent Singularity/Apptainer .sif image can be built from the same recipe). For quantmsdiann to select a locally built image automatically, it must be tagged with the prefix ghcr.io/bigbio/diann: followed by the release tag (e.g. ghcr.io/bigbio/diann:2.5.1 or ghcr.io/bigbio/diann:2.5.1-enterprise); the workflow then resolves the container for the requested version through the corresponding diann_v*X*_*Y*_*Z* profile selected at runtime.

Supplementary Note 3. Identification filtering and protein counting

All identification counts reported in the cell-line reanalyses and pan-cohort (main-text Figs. 3 and 4 and Figs. S4–S13 below) are target-only under a conservative contaminant/decoy filter: any row whose Protein.Group field contains a token prefixed with CONTAM_, Cont_, ENTRAP_, or DECOY_ is dropped before counting, including mixed groups. Original-paper headline counts were taken from the corresponding publication's table or text without re-filtering; where the original applied a per-protein peptide threshold (for example, the ProCan-DepMapSanger paper reports both an 8,498 protein-group headline at Global.Q.Value ≤ 0.01 and a 6,692 count under the additional ≥ 2 supporting-peptide rule), the like-for-like quantmsdiann count is reported under the same threshold.

Supplementary Note 4. Per-process runtime and resource envelopes

These figures decompose the end-to-end wall-clock and resource use reported in main-text Fig. 2b by Nextflow process, derived from trace.txt records collected across the ProteoBench benchmark cohorts and the PXD071075 single-cell production sweep.

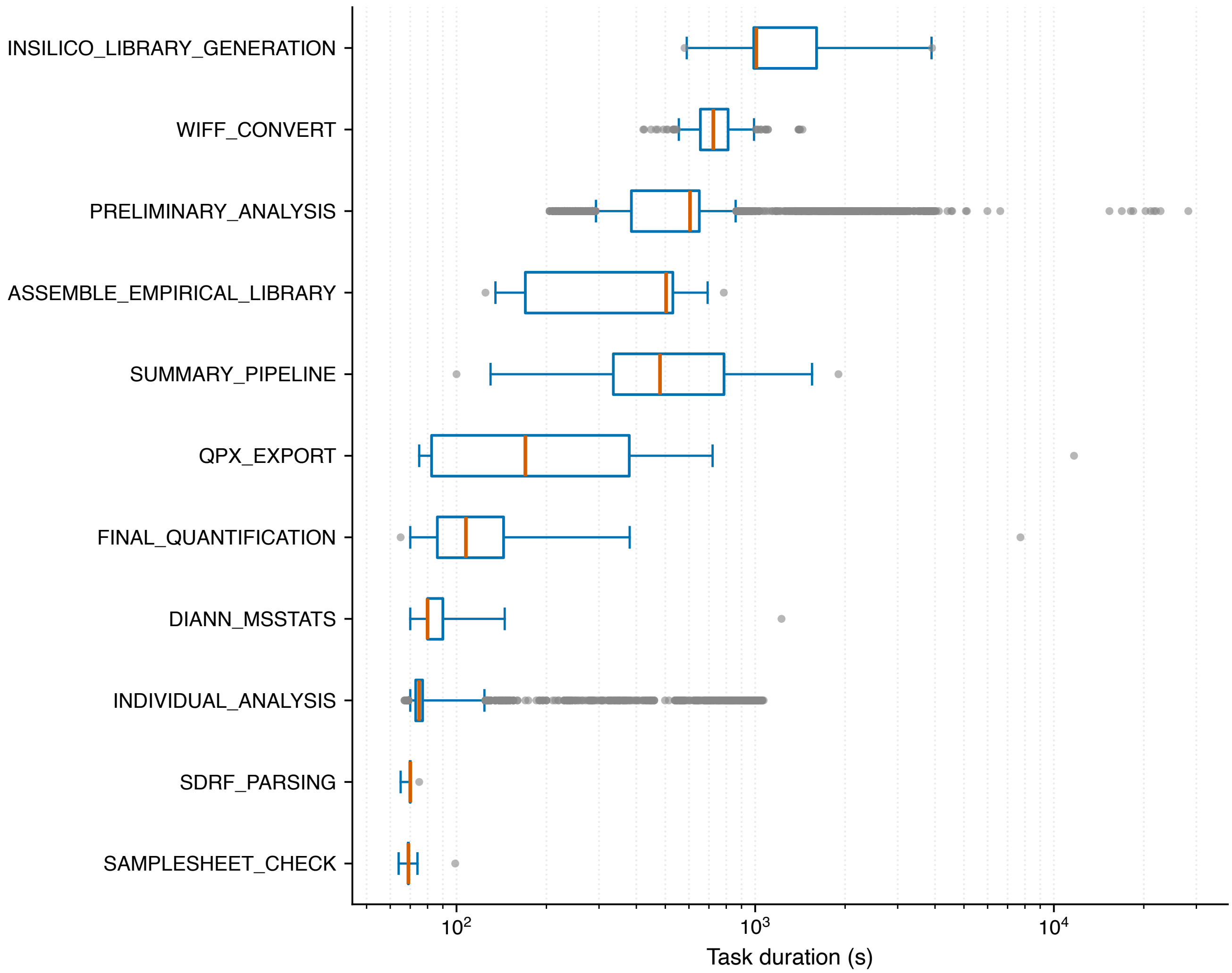


Figure S1: **Per-step wall-clock distribution across Nextflow processes.** Distribution of task wall-clock time (boxplot, median and IQR) for each `quantmsdiann` process across the ProteoBench benchmark cohorts and the PXD071075 cluster-node sweep. The per-file parallel stages (`FILE_PREPARATION`, `PRELIMINARY_ANALYSIS`, `INDIVIDUAL_ANALYSIS`) run as one task per raw file and determine the workflow's parallelisation envelope, while the collective stages (`INSILICO_LIBRARY_GENERATION`, `ASSEMBLE_EMPIRICAL_LIBRARY`, `FINAL_QUANTIFICATION`, `DIANN_MSSTATS`) run once per cohort and set the wall-clock floor at maximum cluster width.

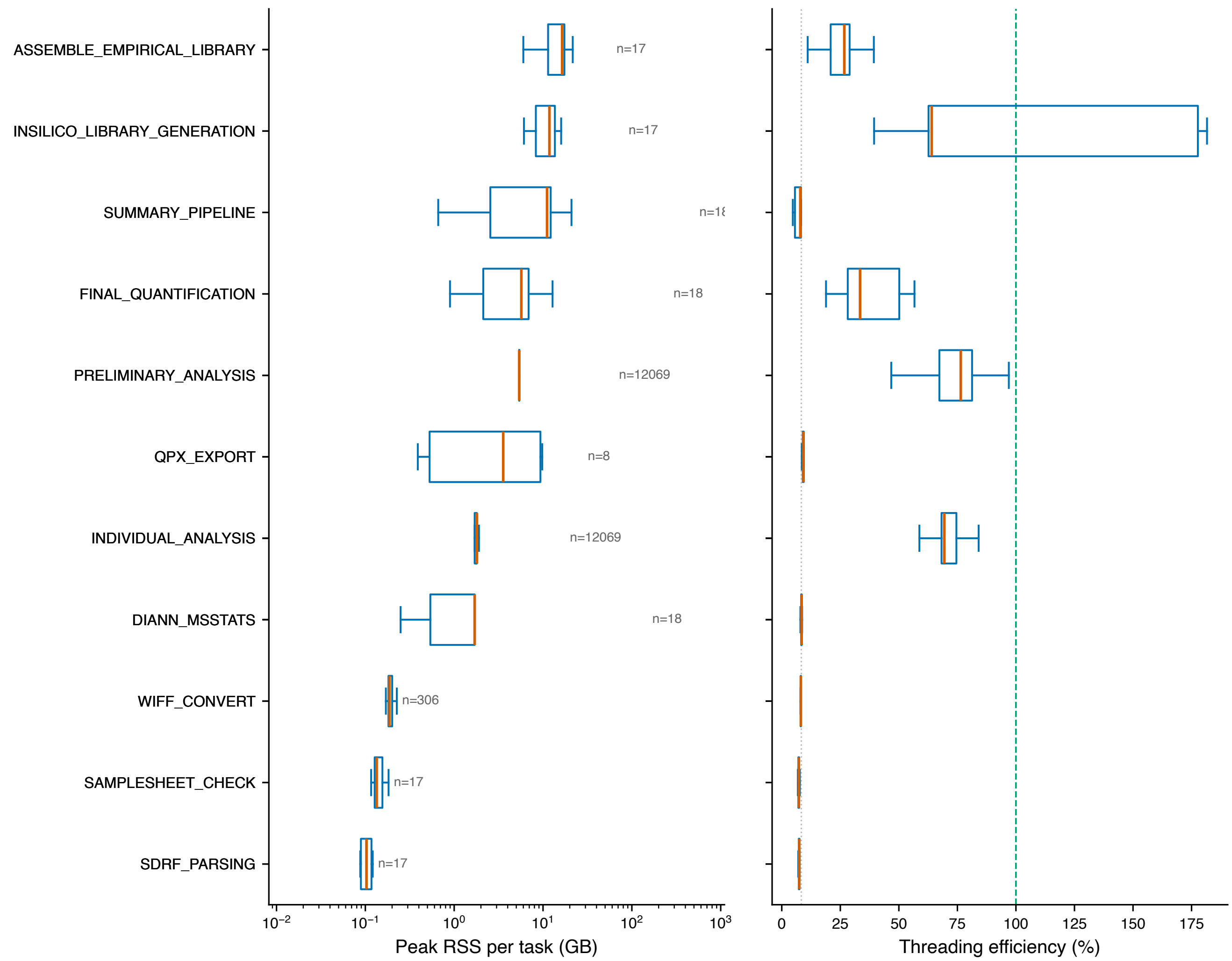
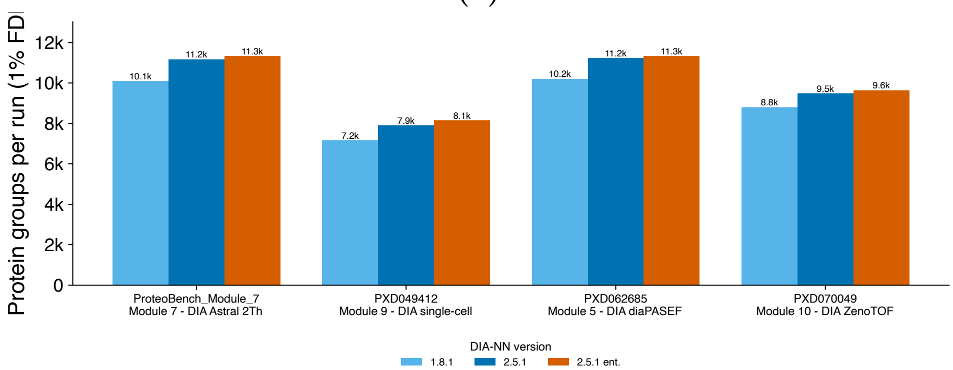


Figure S2: **Peak memory and CPU envelope per Nextflow process.** Per-task peak resident-set-size memory (left panel) and CPU-hours (right panel) per `quantmsdiann` process across the ProteoBench benchmark cohorts and the PXD071075 cluster-node sweep. The collective `FINAL_QUANTIFICATION` stage dominates memory use on large cohorts; per-file stages remain well within the commodity 8–16 GB per-task envelope typical of HPC partitions.

Supplementary Note 5. ProteoBench protein-group depth across `DIA-NN` versions

The main text reports ProteoBench quantification accuracy (Fig. 2c); the identification depth behind the same runs is summarised here. Precursors identified in at least one run rise by 14–23% from `DIA-NN` 1.8.1 to 2.5.1-enterprise on all four modules (largest on timsTOF dia-PASEF, 97,149 → 119,726); proteins quantified in every run rise by up to 20% (single-cell Astral, 6,124 → 7,349). Protein groups are counted from the per-precursor `DIA-NN` report at a 1% run-specific protein-group q-value, target-only, as the average per run (panel a) and the number quantified in every run ("complete profiles", panel b).

(a)



(b)

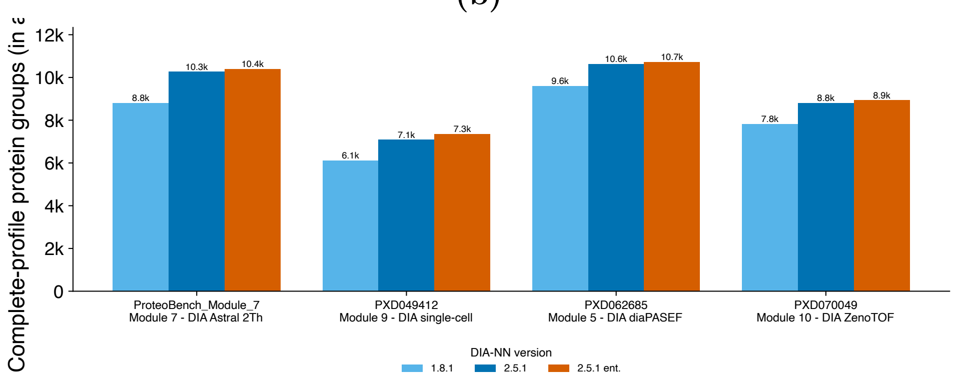


Figure S3: **Protein-group depth on ProteoBench across three `DIA-NN` configurations and four DIA modalities.** (**a**) Protein groups per run (averaged over the six replicates) at a 1% run-specific protein-group FDR, by `DIA-NN` configuration (1.8.1 and 2.5.1 light to dark blue; 2.5.1-enterprise in amber) from one `quantmsdiann` invocation. (**b**) Complete-profile protein groups (quantified in every run). Both metrics rise monotonically from 1.8.1 to 2.5.1 to the enterprise build on every module; the gain is larger for the complete profiles (up to +20% to the enterprise build) than for the per-run average (+8–14%), reflecting improved cross-run consistency in the newer release.

Supplementary Note 6. Per-cohort completeness and overlap (cell-line reanalyses)

These figures complement main-text Fig. 4 with per-condition completeness, per-protein peptide counts, gene/accession overlap, and per-cell-line missingness for each reanalysed cohort.

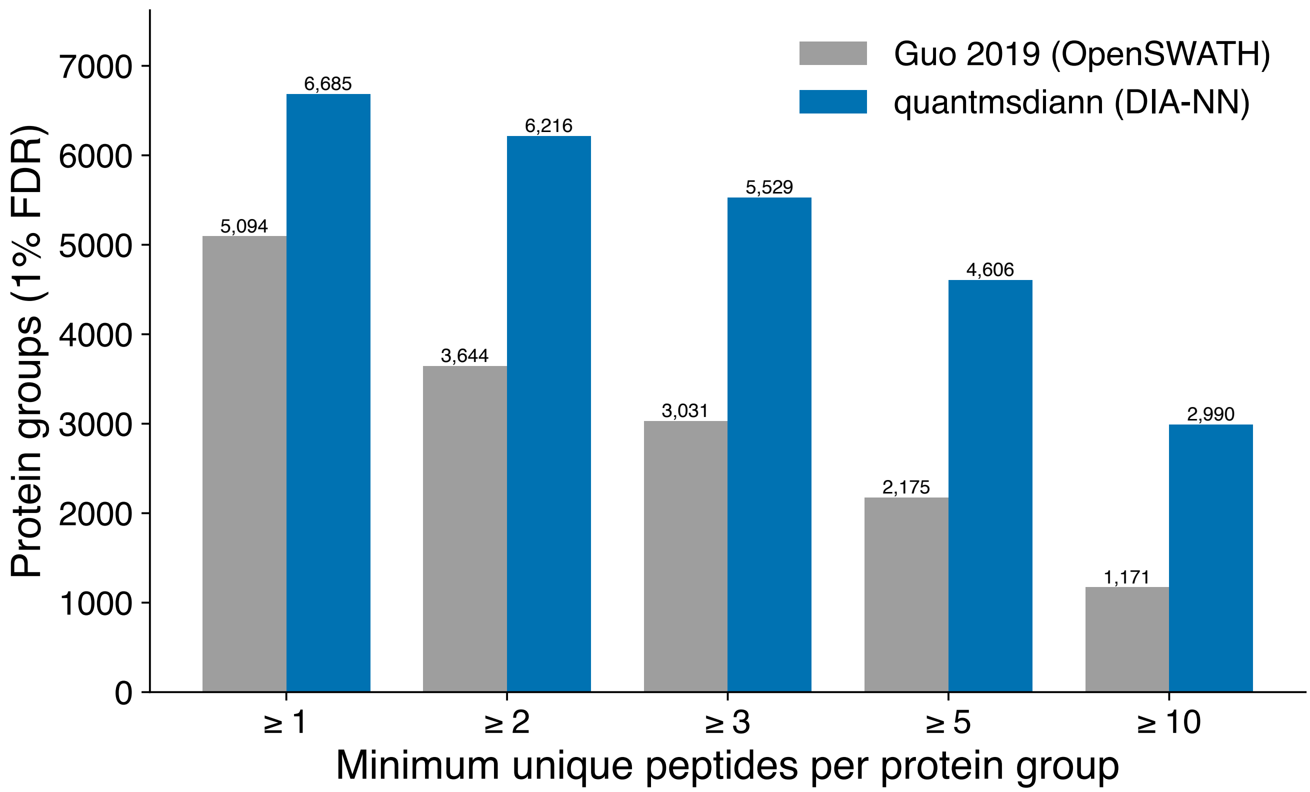


Figure S4: **PXD003539: distribution of peptides per protein group.** `quantmsdiann` (DIA-NN, blue) versus Guo 2019 OpenSWATH (grey) on the same NCI-60 raw data.

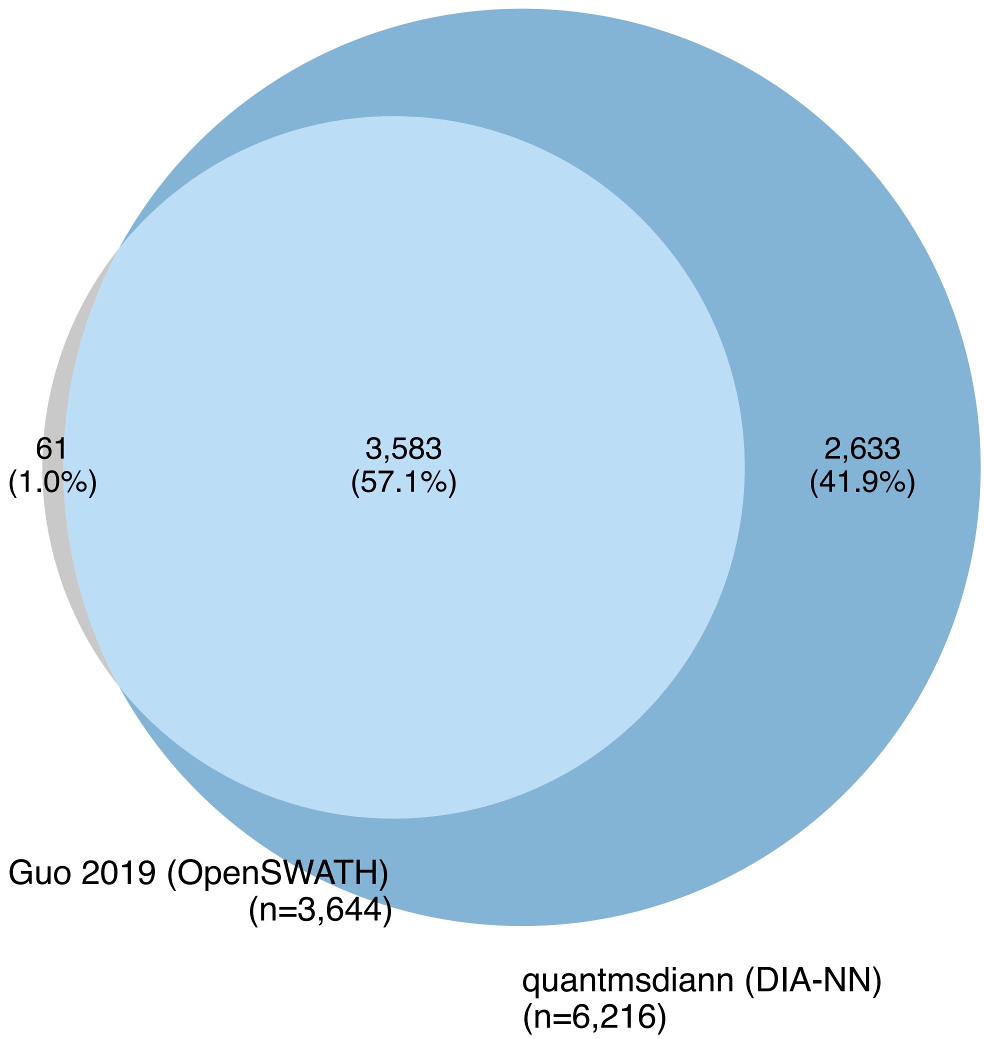


Figure S5: **PXD003539: UniProt-accession overlap between original OpenSWATH analysis and `quantmsdiann` reanalysis.**

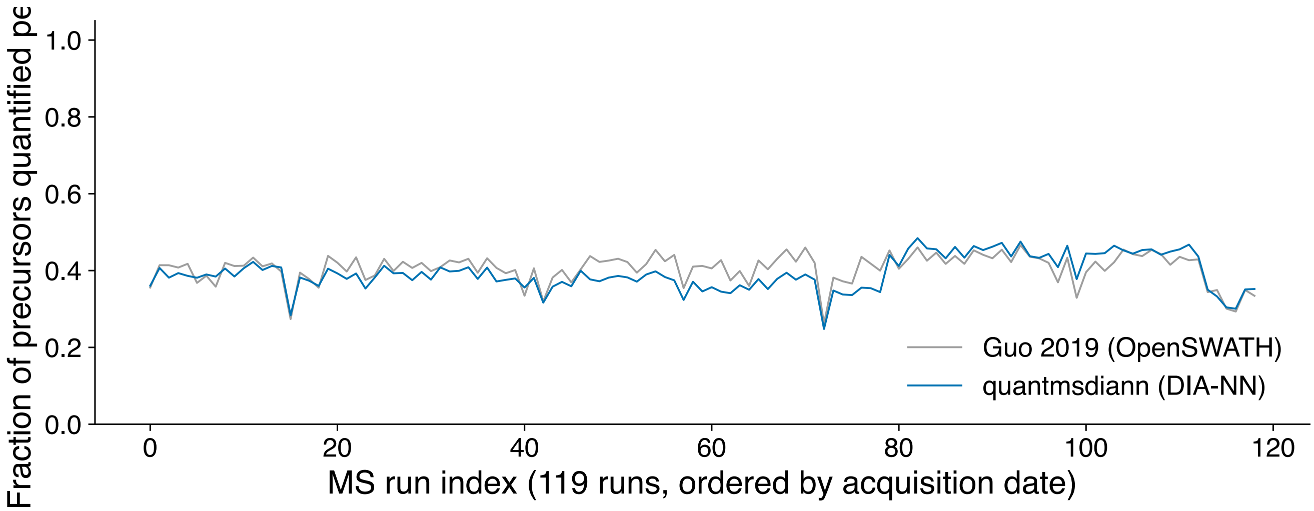


Figure S6: **PXD003539: per-run missing-value fraction.** Distribution of per-run missingness across the NCI-60 panel for both pipelines.

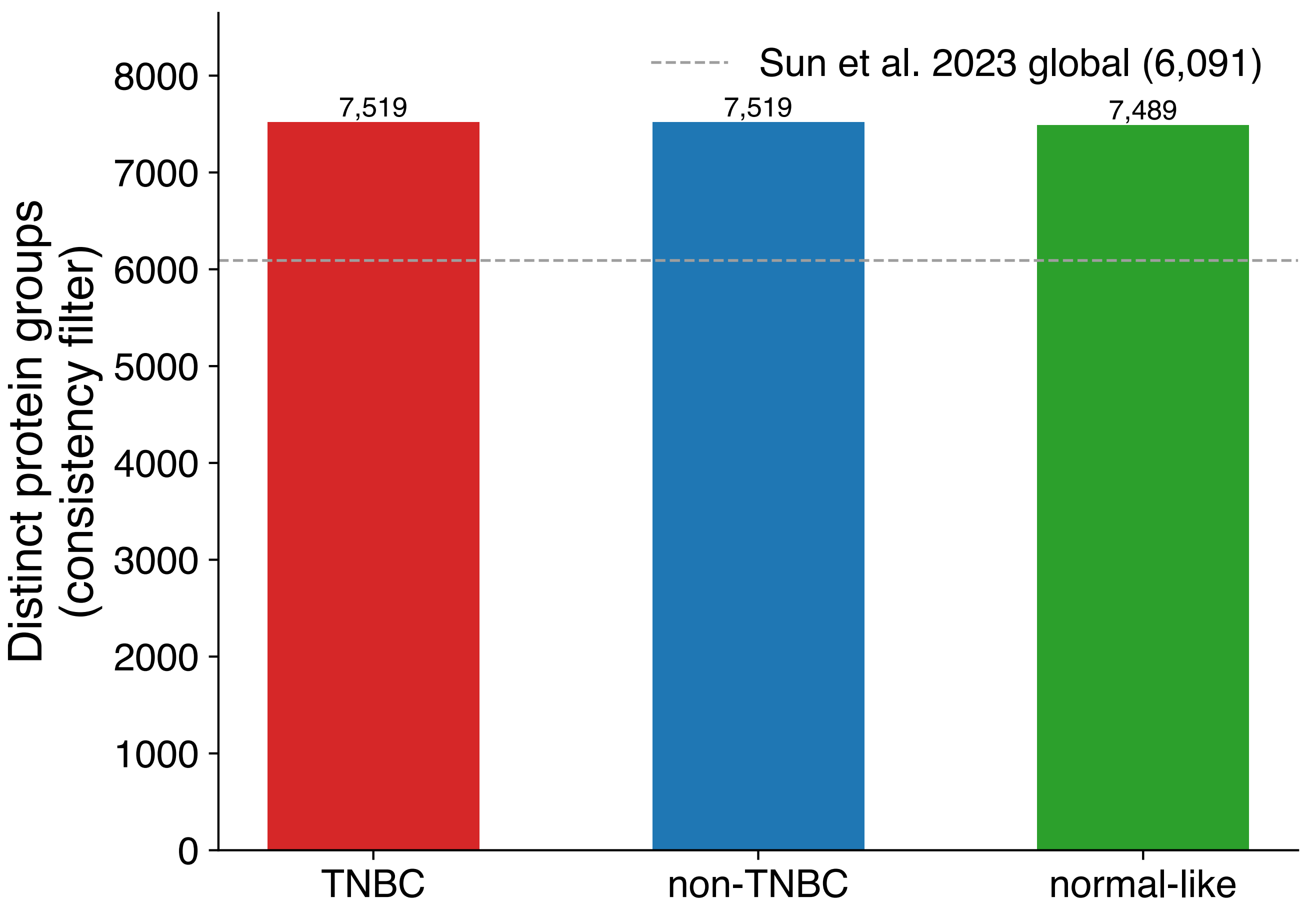


Figure S7: **PXD004701 (Sun 2023): protein groups per breast-cancer subtype.** Distribution of identifications across the 76 cell lines stratified by TNBC / non-TNBC subtype.

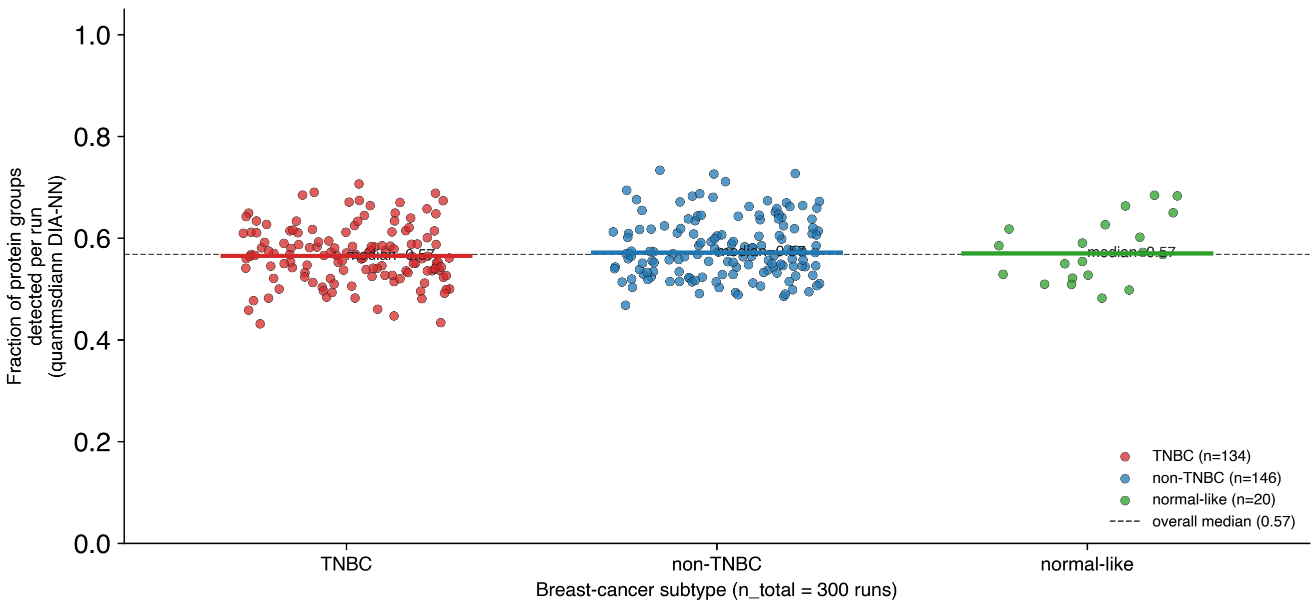


Figure S8: **PXD004701 (Sun 2023): per-run missing-value fraction.**

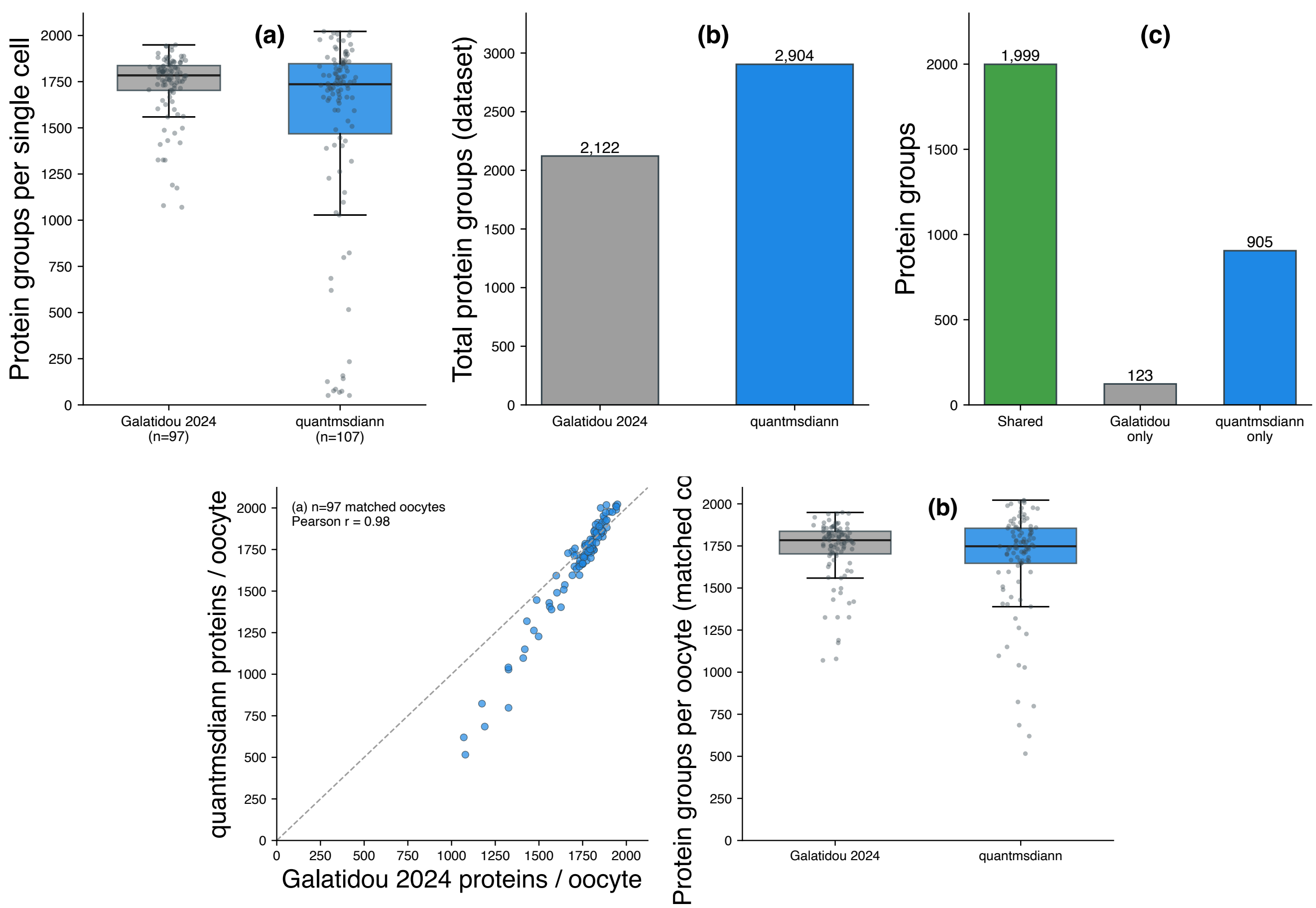


Figure S9: **MSV000093870 (Galatidou 2024, single-cell oocyte plexDIA): `quantmsdiann` reanalysis versus the original.** (**a**) Protein groups per single cell (Galatidou $n = 97$ post-QC oocytes; `quantmsdiann` $n = 107$ channel-confident cells): comparable median depth (1,784 original versus 1,736 `quantmsdiann`). (**b**) Total protein groups across the dataset: 2,122 (original) versus 2,904 (`quantmsdiann`). (**c**) Protein-group overlap at the leading-protein level (like-for-like): 1,999 shared, 123 original-only, 905 `quantmsdiann`-only, i.e. 94% of the originally reported protein groups recovered (1,999 of 2,122) plus 905 detected only by `quantmsdiann`. (Expanding `quantmsdiann`'s protein groups to all member accessions gives 3,552 accessions, but the overlap here is computed group-to-group so both sides use the same entity.) (**d**) QC-matched per-oocyte comparison: with the two analyses restricted to the same 97 cells, per-cell protein-group depth is equivalent (median 1,784 versus 1,748; Pearson $r = 0.98$). This figure underpins the single-cell panel of the main reanalysis figure.

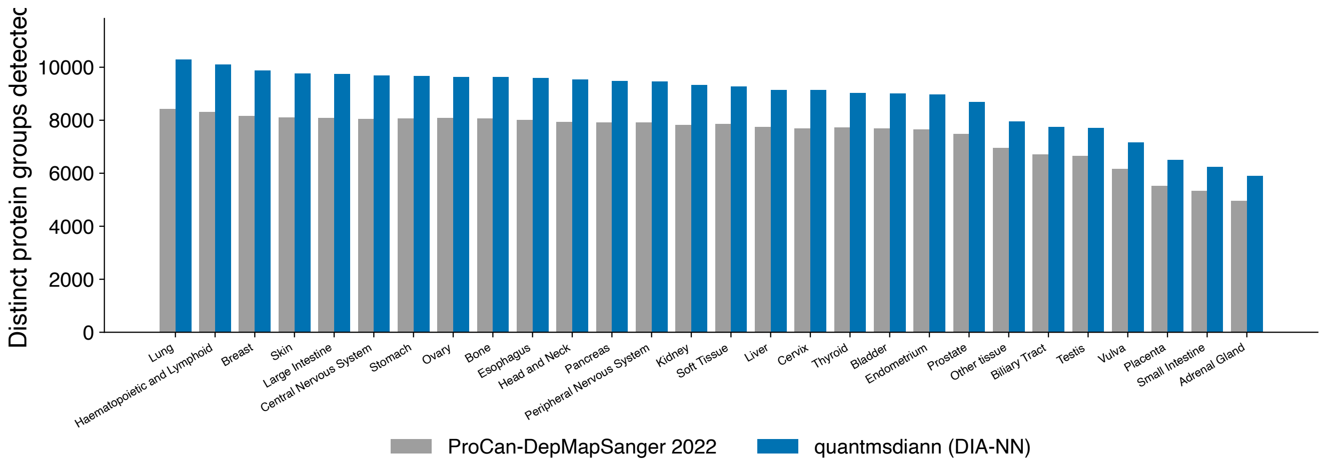


Figure S10: **PXD030304 (ProCan-DepMapSanger): protein groups per tissue.** Distribution of identifications across the 947 cell lines stratified by tissue of origin.

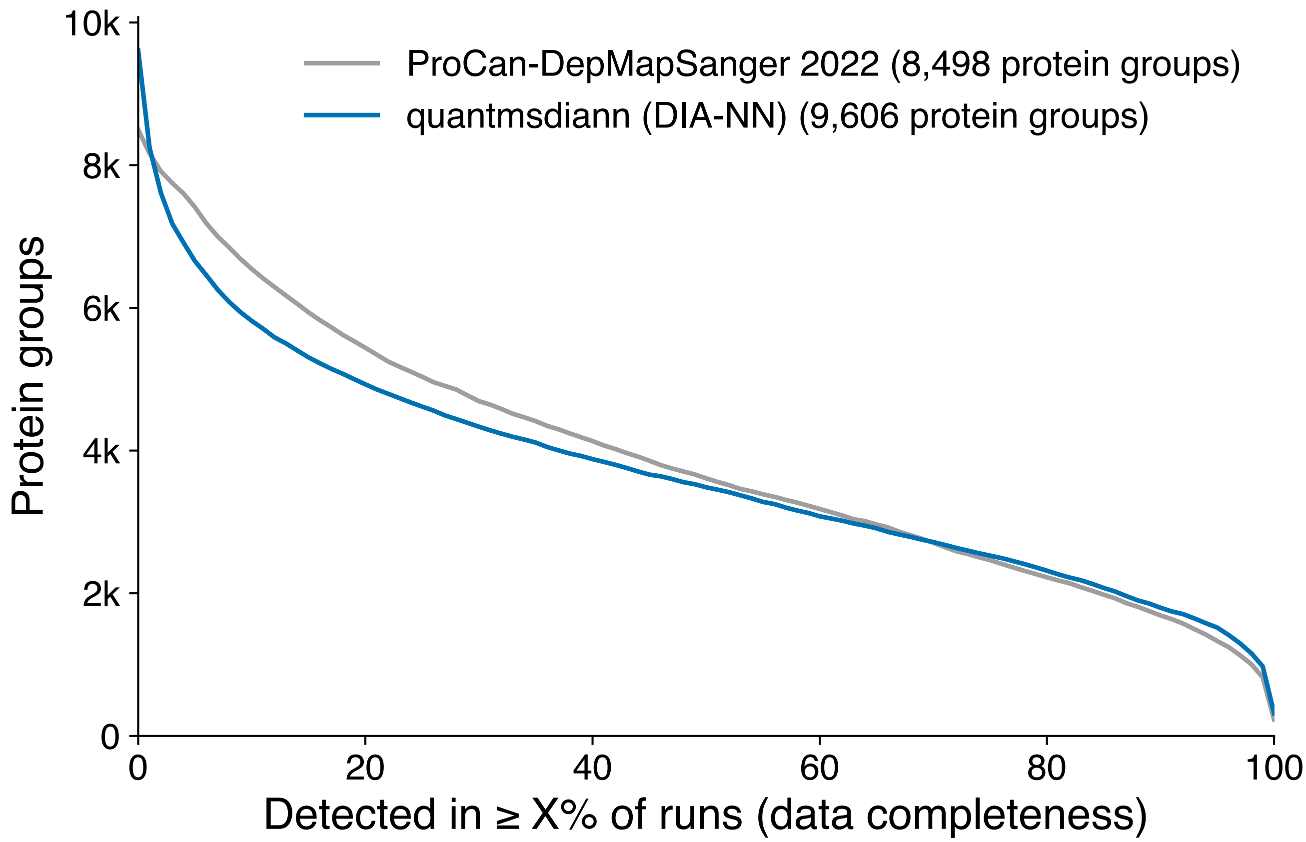


Figure S11: **PXD030304: per-run missing-value fraction across the 947-cell-line panel.**

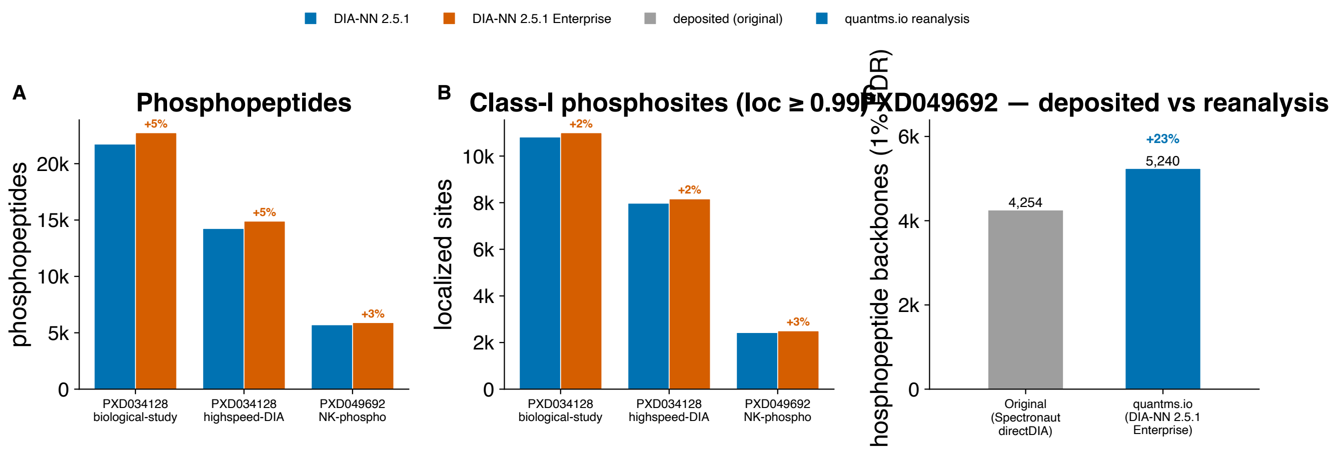


Figure S12: **Phosphoproteomics reanalysis by `quantmsdiann` (DIA-NN, max. two variable modifications per peptide, 1% FDR).** (**a**) Phosphopeptides (distinct phospho-bearing modified precursors) and (**b**) class-I phosphosites (localization probability $\geq$ 0.99, unique by protein and residue) for the phospho-enriched cohorts under standard DIA-NN 2.5.1 versus 2.5.1-enterprise: PXD034128 (HeLa EGF, biological-study and high-speed acquisitions) and PXD049692 (NK-cell Fe-NTA diaPASEF). The enterprise build adds a few percent of identifications across both metrics. (**c**) PXD049692 deposited-versus-reanalysis: distinct stripped phosphopeptide backbones on the identical ten diaPASEF runs, comparing the originally deposited Spectronaut directDIA report with the `quantmsdiann` DIA-NN 2.5.1-enterprise reanalysis. Stripped backbones are the engine-independent metric, as the deposited fragment-level report carries no per-site localization probability; `quantmsdiann` recovers 23% more phosphopeptide backbones (5,240 vs 4,254). The non-enriched PXD034623 (Galectin-1 DIA) is reanalysed for completeness but omitted here, as its ∼280 phosphopeptides are not comparable to the enriched cohorts.

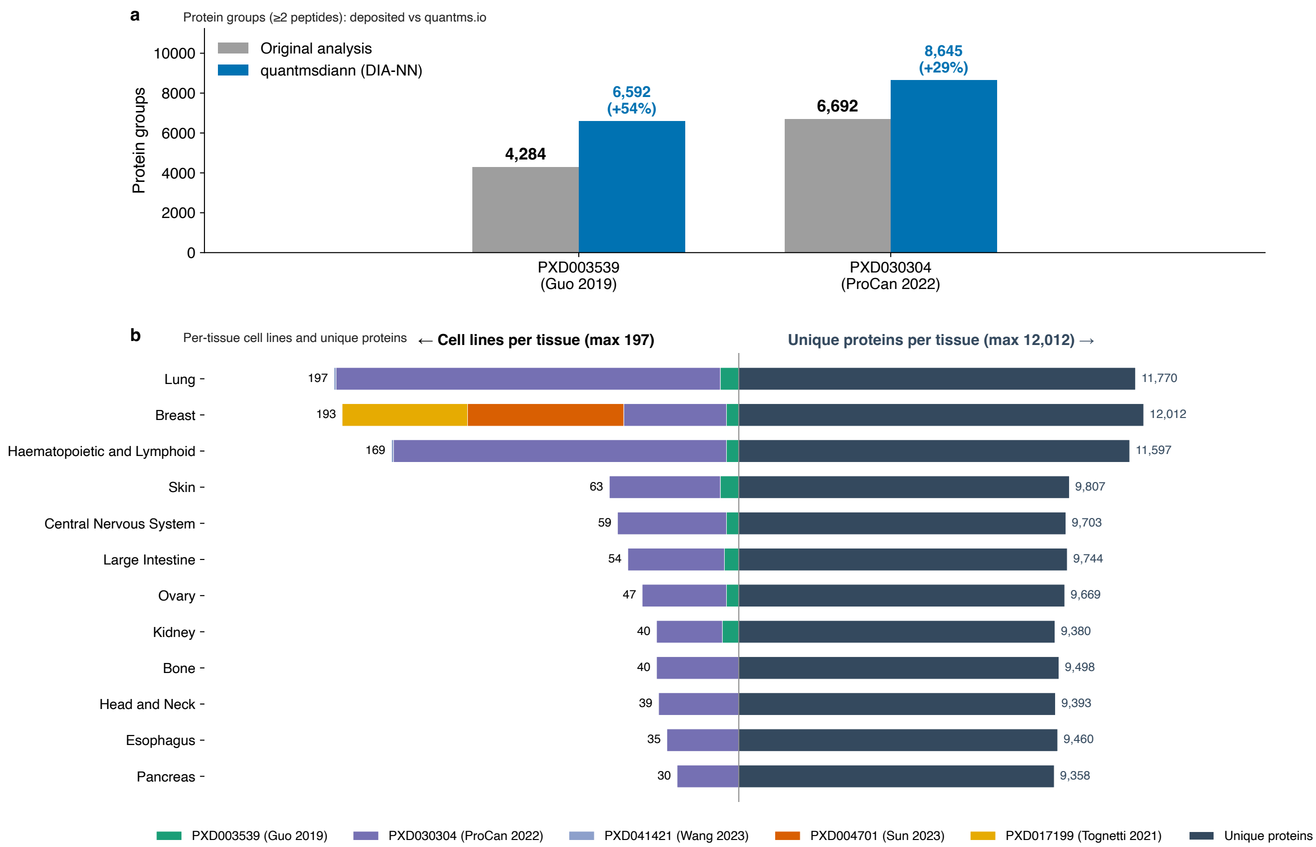


Figure S13: **Pan-cohort of five public DIA cell-line reanalyses by `quantmsdiann` under a single SDRF-driven invocation.** Combines PXD003539 (NCI-60, Guo 2019), PXD030304 (ProCan-DepMapSanger, 947 lines), PXD004701 (Sun 2023, 76 breast-cancer lines), PXD041421 (Wang 2023), and PXD017199 (Tognetti 2021, 67 breast-cancer lines). (**a**) For the two cohorts with a published DIA protein-group headline (NCI-60 and ProCan), protein groups requiring ≥ 2 unique peptides: the originally published count (grey) and the `quantmsdiann` reanalysis (blue), with the same filter applied to both sides (these two comparisons also appear in main-text Fig. 4a). (**b**) Back-to-back per-tissue summary across the combined panel: cell lines contributing to each tissue, stacked by cohort (left), and the union of unique UniProt proteins detected per tissue (right).

Supplementary Note 7. Software and resources

Table S2 lists the software packages, formats, and public resources used by `quantmsdiann` and in this study, with their canonical repositories or portals. `quantmsdiann` and its supporting tools (`sdrf-pipelines`, `pmultiqc`, QPX, `wiffconverter`, container recipes) are released open-source under the `bigbio` organisation; per-version `DIA-NN` container images are built from the recipes in `quantms-containers` (Supplementary Note 2).

Table S2: Software, formats, and public resources used in this study, with their canonical repositories or portals.

Name	Role in this study	Repository / URL
quantmsdiann and the bigbio ecosystem		
`quantmsdiann`	SDRF-driven DIA-NN reanalysis workflow (this work)	`https://github.com/bigbio/quantmsdiann`
quantms	Sibling SDRF-driven multi-engine quantitative proteomics pipeline	`https://github.com/bigbio/quantms`
quantms-containers	Per-release `DIA-NN` container recipes	`https://github.com/bigbio/quantms-containers`
SDRF-Proteomics	Sample-and-Data-Relationship-Format metadata standard	`https://github.com/bigbio/proteomics-sample-metadata`
sdrf-pipelines	SDRF validation and `DIA-NN` configuration generation (`convert-diann`)	`https://github.com/bigbio/sdrf-pipelines`
qpx	Quantitative Proteomics eXchange format and `qpxc` CLI	`https://github.com/bigbio/qpx`
pmultiqc	Proteomics QC reporting (MultiQC extension)	`https://github.com/bigbio/pmultiqc`
wiffconverter	SCIEX `.wiff` to mzML conversion	`https://github.com/bigbio/wiffconverter`
pridepy	PRIDE/MassIVE raw-file download client	`https://github.com/PRIDE-Archive/pridepy`
Workflow engine and analysis tools		
DIA-NN	Data-independent acquisition search and quantification engine	`https://github.com/vdemichev/DiaNN`
Nextflow	Workflow orchestration engine	`https://github.com/nextflow-io/nextflow`
nf-core	Community framework and conventions for the workflow	`https://nf-co.re`
MultiQC	Aggregated analysis-report framework (extended by pmultiqc)	`https://github.com/MultiQC/MultiQC`
MSstats	Statistical quantification; target output format	`https://github.com/Vitek-Lab/MSstats`
ProteoBench	Community benchmarking platform for DIA workflows	`https://github.com/Proteobench/ProteoBench`
ThermoRawFileParser	Thermo `.raw` to mzML conversion	`https://github.com/compomics/ThermoRawFileParser`
Containers and data structures		
Docker	Container runtime (workflow profile)	`https://www.docker.com`
Apptainer (Singularity)	HPC container runtime (workflow profile)	`https://apptainer.org`
BioContainers	Source of the default `DIA-NN` 1.8.1 image	`https://biocontainers.pro`
DuckDB	In-process engine for querying the QPX Parquet archive	`https://github.com/duckdb/duckdb`
MuData (`.h5mu`)	Multimodal container for single-cell export	`https://github.com/scverse/mudata`

Name	Role in this study	Repository / URL
scverse	Single-cell analysis ecosystem (downstream of MuData export)	`https://scverse.org`
Data resources and ontologies		
PRIDE Archive	Source repository for deposited datasets	`https://www.ebi.ac.uk/pride`
ProteomeXchange	Data-deposition consortium / dataset search API	`https://www.proteomexchange.org`
MassIVE	Source repository for the plexDIA cohort	`https://massive.ucsd.edu`
UniProt	Reference protein sequences (search FASTA)	`https://www.uniprot.org`
Cellosaurus	Cell-line nomenclature and tissue annotation	`https://www.cellosaurus.org`

References

[1] S. Kamatchinathan, S. Hewapathirana, C. Bandla, S. Insua, J. A. Vizcaíno, Y. Perez-Riverol, pridepy: a Python package to download and search data from PRIDE database, J Open Source Softw 10 (107) (2025) 7563. doi:10.21105/joss.07563.

[2] E. W. Deutsch, N. Bandeira, Y. Perez-Riverol, V. Sharma, J. J. Carver, L. Mendoza, D. J. Kundu, C. Bandla, S. Kamatchinathan, S. Hewapathirana, et al., The ProteomeXchange consortium in 2026: making proteomics data FAIR, Nucleic Acids Res 54 (D1) (2026) D459–D469. doi:10.1093/nar/gkaf1146.

[3] F. da Veiga Leprevost, B. A. Grüning, S. Alves Aflitos, H. L. Röst, J. Uszkoreit, H. Barsnes, M. Vaudel, et al., BioContainers: an open-source and community-driven framework for software standardization, Bioinformatics 33 (16) (2017) 2580–2582. doi:10.1093/bioinformatics/btx192.